SHREYA SHUKLA

Computer Science Student Specializing in IoT

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PROFESSIONAL SUMMARY

Computer Science student specializing in IoT with strong foundations in programming, data structures, and AI/ML. Experienced in developing innovative projects and participating in hackathons, with a passion for building intelligent systems that integrate software, AI, and connected devices.

TECHNICAL SKILLS

Languages: C, C++, Python, Java
Frameworks: Flask, NodeJS, React
Computer Vision: YOLOv8, OpenCV, MediaPipe
Developer Tools: Git, GitHub, Docker, Postman, Vercel, Render
Data Visualization: MS Excel, MS Word, PowerPoint
Core Computer Science: Data Structures & Algorithms, Operating Systems, DBMS, Networking

EDUCATION

Vellore Institute of Technology, Vellore **2024 – 2028**
Computer Science and Engineering, Internet of Things — CGPA: 9.18/10

City Montessori School (ISC) **2021 – 2023**
12th with Science — Percentage: 97% — Lucknow, Uttar Pradesh

PROJECTS

Near-Miss Safety Dashboard — AI-Based Collision Risk Detection System

Python, YOLOv8, OpenCV, Flask, JavaScript, Leaflet.js

- Engineered a real-time traffic collision risk detection system using YOLOv8, achieving ~25 FPS inference on video streams
- Designed end-to-end computer vision pipeline: frame extraction → object detection → multi-object tracking → Time-to-Collision (TTC) computation → dynamic severity scoring
- Developed Flask-based REST API to stream structured risk events (JSON) for real-time dashboard consumption
- Integrated Open-Meteo API to incorporate live weather conditions into contextual risk assessment
- Built geo-mapped visualization dashboard covering 50km risk zones with interactive event tracking
- Reduced overall inference latency by ~30% through frame batching and processing optimizations

SignSOS — Emergency Sign Language Calling System

Python, Flask, MediaPipe, Scikit-learn, JavaScript, PWA

- Built a real-time hand gesture recognition system translating 13+ ASL gestures into emergency triggers with ~92% classification accuracy
- Designed end-to-end ML pipeline: MediaPipe feature extraction → 42D feature vector → Scikit-learn classifier → Flask REST API
- Optimized inference latency to <150ms per frame, enabling smooth browser-based interaction
- Integrated live GPS tracking and automated WhatsApp emergency alerts via API integration
- Developed Progressive Web App (PWA) with offline caching, OTP-based authentication, and geo-based emergency routing
- Deployed cloud-hosted backend on Render with RESTful architecture supporting concurrent API requests

VITAsk — NLP-Based Intelligent Campus Assistant

Python, NLP, Flask, JSON, Machine Learning

- Developed a domain-specific NLP chatbot to automate campus-related query resolution across academic and administrative domains
- Designed modular intent classification pipeline using structured JSON-based intent mapping with pattern matching and dynamic response routing
- Implemented context-aware response generation for dean, department, and placement-specific queries
- Architected scalable backend in Flask enabling seamless addition of new intents without modifying core routing logic
- Optimized response processing to achieve sub-200ms latency for real-time interaction

KEY ACHIEVEMENTS

- Solved 200+ DSA problems across LeetCode and GeeksforGeeks (GFG)
- Strong in arrays, graphs, dynamic programming (DP), binary search, and bit manipulation
- Regular participant in coding contests